

THE KUBERNETES 100

Ultimate CLI Cheatsheet for Senior Cloud Architects & DevOps

1. CLUSTER INFO & CONTEXT MANAGEMENT

Command	Purpose
<code>kubectl cluster-info</code>	Display addresses of the master and services.
<code>kubectl config view</code>	Show merged kubeconfig settings.
<code>kubectl config get-contexts</code>	List all available cluster contexts.
<code>kubectl config use-context <name></code>	Switch current context to a different cluster.
<code>kubectl config set-context --current --namespace=<ns></code>	Permanently switch default namespace for current context.
<code>kubectl api-resources</code>	List all supported API resource types (great for finding short names).
<code>kubectl api-versions</code>	Print supported API versions on the server.
<code>kubectl get events -A --sort-by='.metadata.creationTimestamp'</code>	ESSENTIAL Global chronological log of all cluster events.

2. NODE DIAGNOSTICS & MANAGEMENT

Command	Purpose
<code>kubectl get nodes -o wide</code>	List nodes with OS, container runtime, and internal/external IPs.
<code>kubectl describe node <node></code>	View node capacity, allocatable resources, taints, and running pods.
<code>kubectl top nodes</code>	Display CPU and Memory usage per node (requires Metrics Server).
<code>kubectl cordon <node></code>	Mark node as unschedulable (prevents new pods from landing).
<code>kubectl uncordon <node></code>	Mark node as schedulable again.
<code>kubectl drain <node> --ignore-daemonsets --force</code>	PRO TIP Gracefully evict all pods for maintenance or AKS upgrades.

Command	Purpose
<code>kubectl taint node <node> key=value:NoSchedule</code>	Apply a taint to repel pods without matching tolerations.
<code>kubectl label node <node> disktype=ssd</code>	Add a label to a node for nodeSelector targeting.

3. POD OPERATIONS & DEEP DEBUGGING

Command	Purpose
<code>kubectl get pods -A</code>	List all pods across all namespaces.
<code>kubectl get pods -o wide</code>	See which node each pod is running on and its IP.
<code>kubectl get pods --show-labels</code>	List pods along with their attached labels.
<code>kubectl get pods -l app=nginx</code>	Filter pods by a specific label selector.
<code>kubectl describe pod <pod></code>	Inspect pod configuration, volume mounts, and scheduling events.
<code>kubectl logs <pod></code>	Print standard output logs for a pod.
<code>kubectl logs -f <pod></code>	Stream logs continuously (tailing).
<code>kubectl logs <pod> -c <container></code>	Print logs for a specific container in a multi-container pod.
<code>kubectl logs -p <pod></code>	INTERVIEW GOLD Get logs from a previously crashed instance of a container.
<code>kubectl exec -it <pod> -- /bin/sh</code>	Open an interactive shell inside a running pod.
<code>kubectl top pods -n <namespace></code>	Check pod CPU/Memory consumption to identify leaks.
<code>kubectl port-forward pod/<pod> 8080:80</code>	Tunnel local port 8080 to pod port 80 for testing.
<code>kubectl delete pod <pod> --grace-period=0 --force</code>	WARNING Forcefully terminate a pod stuck in Terminating state.
<code>kubectl run temp --image=busybox -it --rm -- restart</code>	Spin up a temporary ephemeral pod for network testing (curl/ping).
<code>kubectl cp <pod>:/path/to/remote /path/to/local</code>	Copy files/directories from a pod to your local machine.

4. DEPLOYMENTS & SCALING

Command	Purpose
<code>kubectl get deploy</code>	List deployments and check Ready/Up-to-date counts.
<code>kubectl describe deploy <deploy></code>	Check deployment strategy, replica limits, and active ReplicaSets.
<code>kubectl scale deploy <deploy> --replicas=3</code>	Manually scale the number of pods in a deployment.
<code>kubectl rollout status deploy/<deploy></code>	Watch the status of a rolling update in real-time.
<code>kubectl rollout history deploy/<deploy></code>	View previous deployment revisions.
<code>kubectl rollout undo deploy/<deploy></code>	PANIC BUTTON Revert to the last known working deployment state.
<code>kubectl rollout restart deploy/<deploy></code>	Force a rolling restart of pods (useful for ConfigMap updates).
<code>kubectl set image deploy/<deploy> <container>=<image></code>	Hot-swap a container image via CLI.
<code>kubectl set resources deploy/<deploy> -c=<container> --limits=cpu=200m,memory=512Mi</code>	Update resource limits on the fly.
<code>kubectl autoscale deploy <deploy> --min=2 --max=10 --cpu-percent=80</code>	Create an HPA (Horizontal Pod Autoscaler) automatically.

Command	Purpose
<code>kubectl get rs</code>	List ReplicaSets (managed by deployments).
<code>kubectl delete deploy <deploy></code>	Delete a deployment and all its managed pods.

5. SERVICES & NETWORK ROUTING

Command	Purpose
<code>kubectl get svc</code>	List services and their ClusterIP/ External-IPs.
<code>kubectl describe svc <svc></code>	Check service selectors and routing targets.
<code>kubectl get endpoints <svc></code>	INTERVIEW GOLD Verify if a Service is actually routing traffic to matching pods.
<code>kubectl expose deploy <deploy> --port=80 --target-port=8080</code>	Create a new Service targeting a deployment.
<code>kubectl get ingress</code>	List external routing rules (Ingress controllers).
<code>kubectl describe ingress <ingress></code>	Check TLS termination and path routing configurations.
<code>kubectl get netpol</code>	List NetworkPolicies restricting pod-to-pod traffic.
<code>kubectl describe netpol <netpol></code>	Review specific ingress/egress firewall rules.
<code>kubectl create svc clusterip my-svc --tcp=5678:8080</code>	Quickly create a ClusterIP service via CLI.
<code>kubectl create svc loadbalancer my-svc --tcp=80:8080</code>	Request a public IP from Azure/ Cloud provider.
<code>kubectl port-forward svc/<svc> 8080:80</code>	Port-forward directly to a service instead of a specific pod.

6. CONFIGMAPS & SECRETS (CONFIGURATION MANAGEMENT)

Command	Purpose
<code>kubectl get cm</code>	List ConfigMaps in the current namespace.
<code>kubectl describe cm <cm></code>	View the plaintext contents of a ConfigMap.
<code>kubectl create cm <cm> --from-literal=key=value</code>	Create a ConfigMap from a direct command line argument.
<code>kubectl create cm <cm> --from-file=config.txt</code>	Create a ConfigMap from an external file.
<code>kubectl get secret</code>	List opaque secrets, TLS certs, and Docker registry credentials.
<code>kubectl describe secret <secret></code>	View secret metadata (contents are hidden by default).
<code>kubectl get secret <secret> -o jsonpath='{.data.key}' base64 --decode</code>	PRO TIP Decode a base64 secret value directly in your terminal.
<code>kubectl create secret generic <secret> --from-literal=pass=123</code>	Create a standard opaque secret.

Command	Purpose
<code>kubectl create secret tls <secret> --cert=path/to/cert.crt --key=path/to/key.key</code>	Create a TLS secret for Ingress controllers.
<code>kubectl create secret docker-registry <name> --docker-username=...</code>	Create credentials to pull images from private ACR/registries.

7. PERSISTENT STORAGE

Command	Purpose
<code>kubectl get pv</code>	List cluster-wide PersistentVolumes.
<code>kubectl describe pv <pv></code>	Check volume capacity, access modes, and reclaim policies.
<code>kubectl get pvc</code>	List PersistentVolumeClaims in the namespace.
<code>kubectl describe pvc <pvc></code>	Check if a PVC is successfully 'Bound' to a PV.
<code>kubectl get sc</code>	List StorageClasses (e.g., Azure Disk, Azure File).
<code>kubectl describe sc <sc></code>	View provisioner details and volume binding modes.
<code>kubectl patch pvc <pvc> -p '{"spec":{"resources":{"requests":{"storage":"10Gi"}}}}'</code>	Dynamically expand PVC storage size (if supported by CSI).

8. SECURITY, RBAC & NAMESPACES

Command	Purpose
<code>kubectl get sa</code>	List ServiceAccounts used by pods for API authentication.

Command	Purpose
<code>kubectl get roles</code>	List namespace-scoped RBAC roles.
<code>kubectl get rolebindings</code>	See which users/SAs are bound to namespace roles.
<code>kubectl get clusterroles</code>	List cluster-wide roles (e.g., cluster-admin).
<code>kubectl auth can-i create deployments --as=system:serviceaccount:default:my-sa</code>	INTERVIEW GOLD Test effective permissions for a specific ServiceAccount.
<code>kubectl get ns</code>	List all namespaces in the cluster.
<code>kubectl create ns <ns></code>	Create a new isolated environment.
<code>kubectl get resourcequotas</code>	Check total CPU/Memory consumption limits for the namespace.
<code>kubectl get limitrange</code>	Check default pod CPU/Memory request/limit constraints.

9. ADVANCED FORMATTING & DECLARATIVE OPS

Command	Purpose
<code>kubectl apply -f manifest.yaml</code>	Create or update resources declaratively.
<code>kubectl apply -f ./dir/</code>	Apply an entire directory of YAML files.

Command	Purpose
<code>kubectl diff -f manifest.yaml</code>	PRO TIP Preview changes before applying a YAML file to production.
<code>kubectl delete -f manifest.yaml</code>	Delete all resources defined in a file.
<code>kubectl get pod <pod> -o yaml</code>	Export a pod's live configuration as YAML.
<code>kubectl get pod <pod> -o json</code>	Export a pod's live configuration as JSON for parsing.
<code>kubectl get pods -o custom-columns=NAME:.metadata.name,IP:.status.podIP</code>	Format output into specific, custom columns.
<code>kubectl explain pod.spec.containers</code>	Built-in documentation. Explains YAML structure and available fields.
<code>kubectl patch deploy <deploy> -p '{"spec":{"paused":true}}'</code>	Pause a deployment to make multiple changes without triggering rollouts.
<code>kubectl replace --force -f manifest.yaml</code>	Force replace a resource (deletes and recreates it).
<code>kubectl apply -k ./dir/</code>	Apply resources using Kustomize overlays.